

WHAT CAN A PERSISTENCE DIAGRAM SEE?

INVARIANCE STRUCTURE AND THE DETECTABILITY OF DATA POISONING

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Motivation

- Network Intrusion Detection Systems (NIDS) are necessary to protect computers from malicious network traffic. **Data poisoning** has the potential to corrupt the model by altering its training dataset.
- The defense must run **before training**: unsupervised, with no trustworthy labels.
- Monkam et al. [1] filter poison topologically — bytes $\rightarrow$ image $\rightarrow$ cubical persistence $\rightarrow$ clustering. **What can that lens see?**

Related Work

- Cohen-Steiner et al. (2005) [3]**. Diagrams are stable: small perturbations give small bottleneck change. The warrant for topology as a detector.
- Monkam, De Lucia & Bastian (2024) [1]**. The pipeline reconstructed here; reports 40–70% of poison removed.
- Ferrara (2025) [2]**. Thm. 2 couples classification error and topological distortion — under hypotheses (linear classifiers, Vietoris–Rips, Frobenius balls) that all fail here. A complementary axis: *which points* to attack, where we ask *which algebra* registers.

Research Question

- Hypothesis**: our low capture rate came from unrealistically weak synthetic poison. **Test**: a surrogate-guided, validity-preserving swap attack.
- Refuted**. Nothing approached 40–70%; the highest capture under any committed configuration was 6.48%. The gap is structural, not a matter of realism.

What determines topological detectability — the attack’s invariance structure, or the filtration’s sensitivities?

Contact

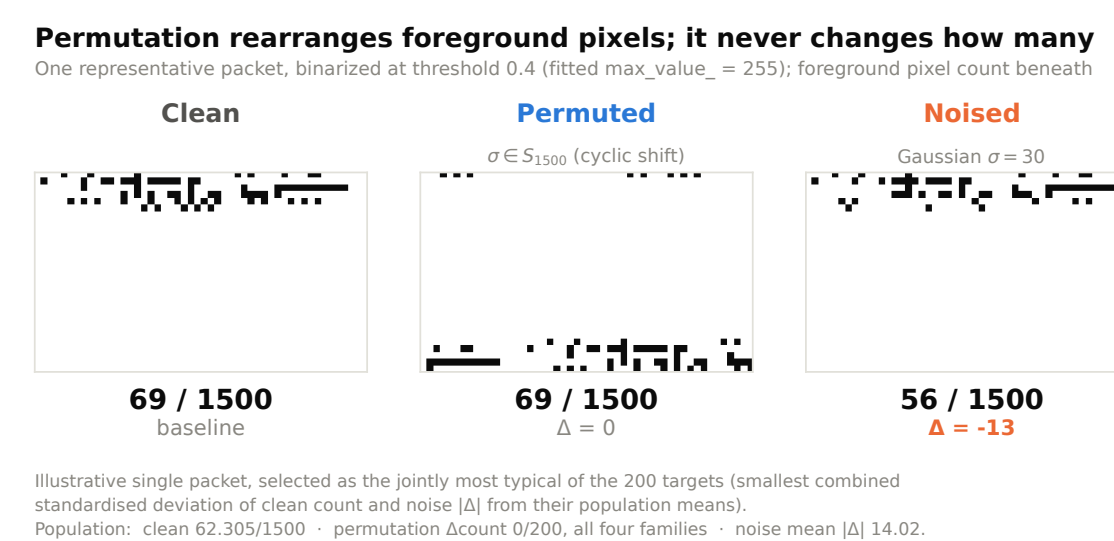
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Methods I: An Invariance Argument

A swap attack applies $\sigma \in S_{1500}$ to the payload $x \in \{0, \dots, 255\}^{1500}$. Binarization b is pointwise, so it commutes with σ :

$$\underbrace{b(x \circ \sigma) = b(x) \circ \sigma}_{\text{equivariance}} \implies \underbrace{\sum_i b((x \circ \sigma)_i)}_{\text{count is invariant}} = \sum_i b(x_i)$$

- Equivariance is not invariance**. Binarization commutes; the *count* is what is fixed. Both steps are needed, and the result holds for every σ , not just the ~ 60 transpositions applied.
- Every count-derived feature is therefore blind to permutation. Detection must flow through the **position-dependent** filtrations (Height, Radial). Invariance that is too coarse is a known hazard — the motivation for group-equivariant operators [4].
- The threshold is a fraction of the batch maximum — data-dependent, but a maximum is itself permutation-invariant.

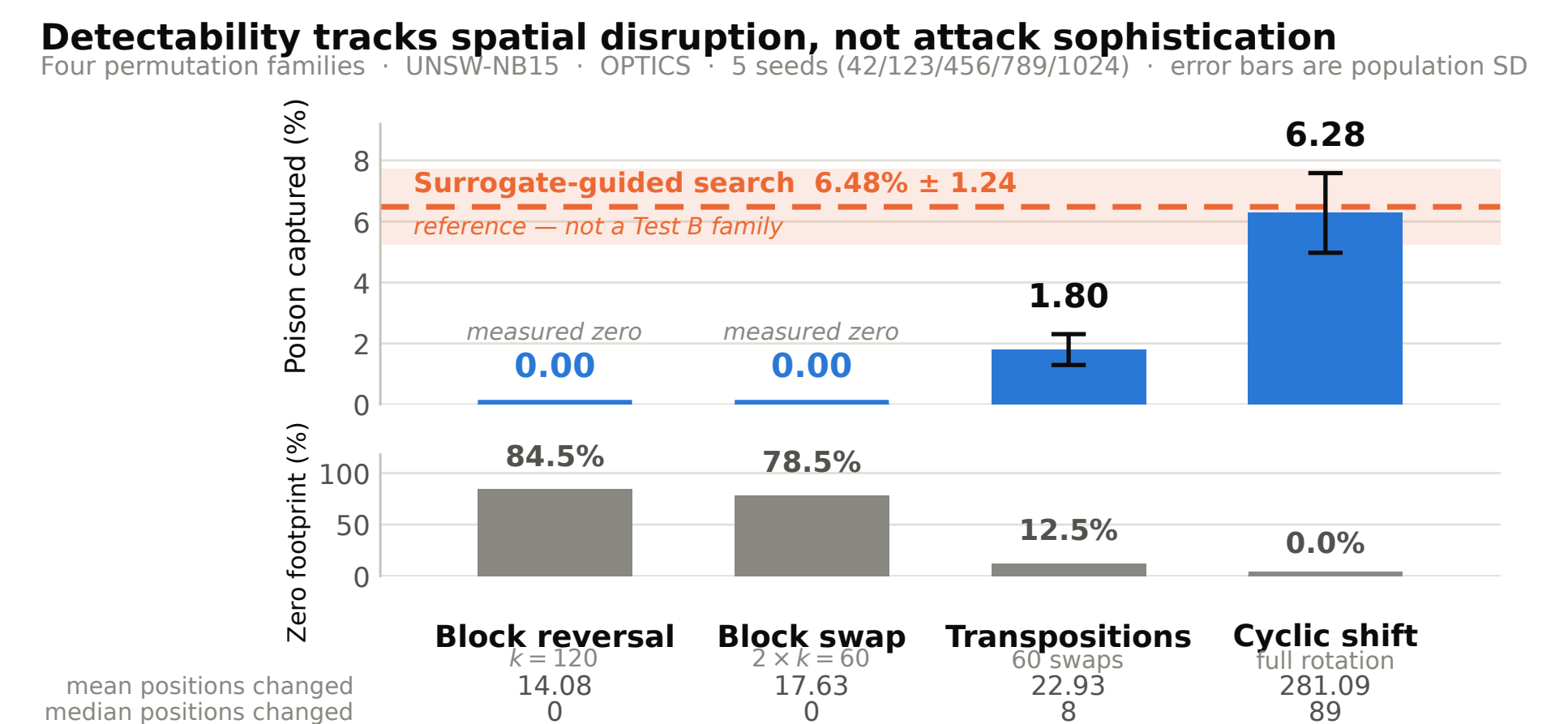


The count channel is provably insensitive to permutation; the position channel stays open.
Attenuated on the foreground-count channel — not absent.

Methods II: Design

- UNSW-NB15, 5,000 packets, 10% poison, 5 seeds. Each 1500-byte payload $\rightarrow 30 \times 50$ image; 5 filtrations $\times 6$ vectorizations $\times 2$ homology dimensions = 60 features — the binarize–filtration–cubical pattern of Garin & Tauzin [5]. OPTICS reported; other clusterers ≈ 0 .
- Capture** = poison landing in 100% *-pure* clusters. Zero-tolerance: a 99%-poison cluster counts for nothing.
- Four multiset-preserving families, so every poisoned packet stays well-formed. Invariance measured directly: 1,600 count checks (200 packets $\times 4$ families $\times 2$ thresholds), Gaussian noise as positive control.

Results: Four Families, One Proof



- The count never moved**: 0/200 for all four families, across 1,600 checks at two thresholds — zero exceptions.
- Identical invariance, a six-point spread in capture**. The invariance class alone does not determine detectability.
- The silent families barely move anything**: median 0 bytes changed; after binarization 88% and 83% of packets are *bit-identical* to their originals, giving identical feature vectors.
- Random swaps are quiet for a computable reason**: foreground fraction $p \approx 0.042$, so only $2p(1-p) \approx 8\%$ of transpositions change the image. Control: noise shifts the count by $|\Delta| = 14.02$.

Discussion

- Capture tracks spatial disruption, not sophistication**: unguided cyclic shift (6.28%) matches surrogate-guided swaps (6.48%).
- A reported negative result**. Guidance gave $+4.68 \pm 1.57$ (5/5 seeds) on UNSW-NB15 but **does not replicate** on CICIDS2017: -0.60 ± 3.03 , 2/5 seeds, per-seed $[-0.4, -2.2, +0.4, +4.2, -5.0]$ — the sign flips and the mean crosses zero. Pre-committed; not rescued.
- The invariance is elementary folklore; we find no prior statement of it for this pipeline. Scope: *multiset-preserving* perturbations. Whether *functional* reorderings in the same group — header order, padding placement — inherit the attenuation is a conjecture this motivates, **not a result**.

Conclusions

- One channel of the detector is **provably insensitive** to permutation attacks. No amount of data can change that.
- Insensitive is not undetectable**: four identically invariant families span 0%–6.3%.
- Detectability is a property of the (attack, detector)

References

- G. F. Monkam, M. J. De Lucia, and N. D. Bastian, “A topological data analysis approach for detecting data poisoning attacks against machine learning based network intrusion detection systems,” *Computers & Security*, vol. 144, art. no. 103929, Sep. 2024.
- M. Ferrara, “Modeling by topological data analysis and game theory for analyzing data poisoning phenomena,” *AIMS Mathematics*, vol. 10, no. 7, pp. 15457–15475, Jul. 2025.
- D. Cohen-Steiner, H. Edelsbrunner, and J. Harer, “Stability of persistence diagrams,” in *Proc. 21st Annu. Symp. Computational Geometry (SCG ’05)*, Pisa, Italy, Jun. 2005,